Pre-Logical States and the Birth of Information: How Reasoning Operates Before Truth Exists

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Abstract

Classical logic—Boolean, fuzzy, modal—assumes that truth values exist before connectives operate on them. But many consequential decisions occur *before truth exists*: a firefighter dispatched to an unidentified highway obstruction, a lawyer confronting a case with no precedent, a city planner choosing infrastructure whose consequences will unfold over decades. We formalize these situations as **pre-logical states** using the four axes of the quaternionic operator framework. A pre-logical state is a point in $V = [0, 1] \times [-1, 1]^3$ where the truth axis r is undefined—not indeterminate (a value exists but is unknown) but genuinely indefinite (no value has crystallized). We prove that the space of possibilities before collapse is a Boolean lattice with exact cardinality, expansion rate, and metric, satisfying all four defining properties of Kauffman's adjacent possible. We establish that the transition from pre-logical to logical states—**collapse**—is governed by a damped harmonic oscillator that is exactly the free energy minimization of Friston's active inference framework, with solenoidal coupling ($\sigma > 0$) necessary for oscillatory behavior. Two independent routes to oscillation—dynamical (solenoidal flow) and geometric (cross-coupled free energy)—are derived. We show that the duality pre-logical/logical generates **information** as its compound concept, connecting the quaternionic framework to Shannon entropy as a special case. Substrate independence is demonstrated across four domains: quantum measurement, legal reasoning, emergency response, and urban planning.

Keywords: pre-logical states; quaternionic logic; adjacent possible; free energy principle; information; collapse; Boolean lattice; substrate independence

1 Introduction

Logic begins with truth values. Boolean logic assigns $\{0, 1\}$; fuzzy logic assigns $[0, 1]$; modal logic quantifies over possible worlds where truth is defined in each. Every formal system in the logical tradition takes truth as *given* and asks what follows from it.

But consider:

- A firefighter receives a report: "large object blocking Highway 45." The object could be a fallen tree, an overturned vehicle, a dead animal, or a boulder. No truth value for "the object is a tree" exists—not because the firefighter doesn't know, but because the situation has not yet been observed. What equipment should the truck carry?
- A lawyer faces a case with no precedent in the jurisdiction. The applicable legal rule is not merely unknown—it does not yet exist. It will be *created* by the judge's ruling. How should the lawyer argue?
- A city planner must allocate budget between water infrastructure and transit for the year 2040. The city's needs in 2040 are not uncertain in the probabilistic sense—they depend on decisions not yet made, by people not yet born, under conditions not yet imaginable.

In each case, the agent must reason *before truth exists*. This is not probabilistic uncertainty (where a truth value exists but is hidden behind a distribution). It is a qualitatively different state: the truth axis is **indefinite**—no value has crystallized.

The quaternionic operator framework [Ornelas Brand, 2026c] provides the formal apparatus to distinguish these states. The operator $O = \{0, 1, +, i, j, k\}$ acts on a space $V = [0, 1] \times [-1, 1]^3$ with four axes: truth (r), potentiality (i), verification (j), and selection (k). The G-lattice of connectives is defined on this space, with the commutation theorem establishing that connectives commute with the group action if and only if they depend solely on r. The companion paper on duality synthesis [Ornelas Brand, 2026d] shows how opposing poles generate new truth through the synthesis operation $a \oplus \bar{a} = \mathrm{lcm}(\Phi(a), \Phi(\bar{a}))$.

This paper addresses what happens *before* the G-lattice can operate—in the regime where r is not yet assigned. We formalize pre-logical states (§3), establish the Boolean lattice structure of the possibility space (§4), define the operations valid before collapse (§5), derive collapse as an algebraic operation (§6), prove the dynamical equivalence between collapse and free energy minimization (§7), identify two independent routes to oscillatory collapse (§8), show that information emerges as the compound concept of the pre-logical/logical duality (§9), and demonstrate substrate independence across four domains (§10).

Contributions.

1. Formal definition of pre-logical states (Definition 1).
2. Proof that the possibility space is a Boolean lattice satisfying all four properties of the adjacent possible (Theorems 5–11).
3. Four operations valid before truth exists (Propositions 14–17).
4. Collapse as an algebraic map $\mathbf{j} : \mathrm{Im}(V) \to V$ (Definition 20, Theorem 22).
5. Exact equivalence between the damped wave and FEP dynamics (Theorem 24).
6. Necessity of solenoidal coupling for oscillation (Remark 25).
7. Two independent routes to oscillation (Proposition 28).
8. Information as compound concept of the pre-logical/logical duality (Proposition 30).
9. Substrate independence across four domains (Section 10).

2 Preliminaries

We recall the necessary elements from the quaternionic operator framework [Ornelas Brand, 2026c] and the prime algebra [Ornelas Brand, 2026a].

The quaternionic operator. The operator $O = \{0, 1, +, i, j, k\}$ acts on $V = [0, 1] \times [-1, 1]^3$, where each point $v = (r, i, j, k) \in V$ represents:

- $r \in [0, 1]$: truth (real axis),
- $i \in [-1, 1]$: potentiality (imaginary axis),
- $j \in [-1, 1]$: verification (empirical axis),
- $k \in [-1, 1]$: recursive selection (decision axis).

The four axes satisfy the quaternionic relation $ij = k$: potentiality composed with verification yields selection.

The G-lattice. The connectives $\{\wedge, \vee, \neg, \to\}$ form a residuated lattice on V with the SU(2) group action $\alpha_q(v) = qvq^{-1}$ for unit quaternions $q \in S^3$. The *commutation theorem* [Ornelas Brand, 2026c] states that a connective c satisfies $c \circ \alpha_q = \alpha_q \circ c$ for all q if and only if c depends only on r. Connectives that touch the imaginary axes are context-sensitive.

Prime algebra. Each concept C is encoded as a product of primes $\Phi(C) = \prod p_i^{a_i}$ drawn from a set of k irreducible attributes [Ornelas Brand, 2026a]. The prime factor count $\omega(\Phi(C))$ measures constructive complexity. The operations gcd (shared structure), lcm (combined structure), and mod (containment test) are the algebraic primitives.

Synthesis. The duality synthesis operation [Ornelas Brand, 2026d] combines opposing poles: $a \oplus \bar{a} = \mathrm{lcm}(\Phi(a), \Phi(\bar{a}))$, producing a compound concept whose complexity strictly exceeds that of either pole when both contribute at least one distinguishing prime.

Free energy principle. Under the Laplace approximation with generalized coordinates $\tilde{\mu} = (\mu, \mu')$, the Friston dynamics [Friston, 2010, Buckley et al., 2017] take the form:

$$\dot{\tilde{\mu}} = D\tilde{\mu} - (\Gamma + Q)\,\nabla_{\tilde{\mu}} F \tag{1}$$

where D is the shift operator, Γ the dissipation matrix, Q the solenoidal (antisymmetric) matrix, and F the variational free energy.

3 The Pre-Logical State

Definition 1 (Pre-logical state)**.** *A **pre-logical state** is a point in the extended space $\hat{V} = \{?\} \cup [0,1] \times [-1,1]^3$ of the form:*

$$v_{\mathrm{pre}} = (?,\ i,\ j,\ k) \quad \textit{with } i > 0 \textit{ and } j < 1 \tag{2}$$

*where $? \notin [0,1]$ denotes that the truth axis is **undefined** —not a missing value (as in incomplete information) but a genuinely non-existent value (nothing has crystallized to assign truth to).*

The conditions $i > 0$ and $j < 1$ are not arbitrary:

- $i > 0$: at least some potentiality remains open. If $i = 0$, there are no possibilities to explore—the state is already determined (the dogma pathology of Ornelas Brand 2026c).
- $j < 1$: verification is incomplete. If $j = 1$, the state has been fully verified and truth is assigned—we are already in the logical regime.

Remark 2 (Indefinite vs. indeterminate)**.** *The distinction between* indefinite *and* indeterminate *is critical. An indeterminate truth value exists but is epistemically inaccessible ("the coin has landed but I haven't looked"). Probability theory handles indeterminacy. An indefinite truth value does not exist ("the coin is still spinning"). No probability distribution can be assigned because the sample space itself is open. The pre-logical state formalizes indefiniteness, not indeterminacy.*

Remark 3 (Relation to the four pathologies)**.** *The pathologies identified in Ornelas Brand [2026c]—dogma ($i = 0$), delirium ($r = 0$), pseudoscience ($j = 0$), and stagnation ($k = 0$)—are collapses of individual axes within the logical regime. The pre-logical state is qualitatively different: it is not a collapse of r to zero (that would be delirium, where truth is denied) but the* absence *of r altogether. The pre-logical state precedes the regime where pathologies are defined.*

4 The Boolean Lattice of Possibilities

Before truth crystallizes, the agent navigates a space of *possibilities*—potential truths that might emerge upon collapse. We show that this space has the structure of a Boolean lattice.

Definition 4 (Prime product space)**.** *Let $P = \{p_1, p_2, \ldots, p_k\}$ be a set of k distinct primes representing irreducible attributes. The **prime product space** is:*

$$\mathcal{L}(P) = \left\{ \prod_{i \in S} p_i : S \subseteq \{1, 2, \ldots, k\} \right\} \tag{3}$$

with partial order given by divisibility: $A \leq B \iff A \mid B$.

Theorem 5 (Boolean lattice)**.** *$(\mathcal{L}(P), |)$ is a Boolean lattice isomorphic to $(2^P, \subseteq)$:*

- Isomorphism*:* $\varphi(S) = \prod_{i \in S} p_i$ *is a bijection (by the Fundamental Theorem of Arithmetic).*
- Meet*:* $\gcd(A, B) = \varphi(\varphi^{-1}(A) \cap \varphi^{-1}(B))$.
- Join*:* $\operatorname{lcm}(A, B) = \varphi(\varphi^{-1}(A) \cup \varphi^{-1}(B))$.
- Complement*:* $\overline{A} = (\prod P)/A$.
- Bottom*:* 1 *(empty product).* Top*:* $\prod P$ *(all primes).*
- Atoms*: individual primes* $p_1, \ldots, p_k$.
- Rank function*:* $r(A) = \omega(A)$ *(number of distinct prime factors).*

Proof. The map $\varphi : 2^P \to \mathcal{L}(P)$ defined by $\varphi(S) = \prod_{i \in S} p_i$ is a bijection by the Fundamental Theorem of Arithmetic. The lattice operations follow from $\min(a_i, c_i)$ and $\max(a_i, c_i)$ on exponents, which correspond to intersection and union of the index sets when exponents are binary. □

Definition 6 (Adjacent possible in the lattice)**.** *Given a set of known concepts $K \subseteq \mathcal{L}(P)$, the* ***adjacent possible*** *is:*

$$AP(K) = \{x \in \mathcal{L}(P) \setminus K : \exists\, y \in K \ \text{ s.t. } d_H(x, y) = 1\} \tag{4}$$

where d_H is the Hamming distance between binary signatures (number of primes by which x and y differ).

We prove that this lattice satisfies all four defining properties of Kauffman's adjacent possible [Kauffman, 2000, Loreto et al., 2017].

4.1 Property 1: Expansion

Theorem 7 (Exact doubling)**.** *Adding one new prime p_{k+1} to the repertoire exactly doubles the space:*

$$|\mathcal{L}(P \cup \{p_{k+1}\})| = 2 \cdot |\mathcal{L}(P)| \tag{5}$$

Proof. $|\mathcal{L}(P)| = |2^P| = 2^k$. Adding p_{k+1} gives $|\mathcal{L}(P \cup \{p_{k+1}\})| = 2^{k+1} = 2 \cdot 2^k$. The new elements are exactly $\{A \cdot p_{k+1} : A \in \mathcal{L}(P)\}$—every existing concept paired with the new prime. The old elements remain. □

Corollary 8. *After discovering n new primes sequentially, the space grows by factor 2^n. Growth is exponential in the number of discoveries.*

4.2 Property 2: Path dependence

Theorem 9 (Exploration depends on history)**.** *The set of explored concepts at time t depends on the order in which concepts are explored, not just on which primes are available.*

Proof. Let $P = \{2, 3, 5\}$ and suppose an agent explores one adjacent concept per step from $K_0 = \{1\}$.

Path α: choose $p = 2$ first. At $t = 1$: $K = \{1, 2\}$, $AP = \{3, 5, 6, 10\}$. Choose $6 = 2 \cdot 3$: $K_\alpha = \{1, 2, 6\}$.

Path β: choose $p = 5$ first. At $t = 1$: $K = \{1, 5\}$, $AP = \{2, 3, 10, 15\}$. Choose $15 = 3 \cdot 5$: $K_\beta = \{1, 5, 15\}$.

Both paths have the same P and the same total space $\mathcal{L}(P) = \{1, 2, 3, 5, 6, 10, 15, 30\}$. But at $t = 2$, $K_\alpha \cap K_\beta = \{1\}$: the explored sets are disjoint except for the bottom element. Their adjacent possibles also differ. □

4.3 Property 3: Faster-than-exploration

Theorem 10 (Vanishing exploration ratio)**.** *For any agent exploring at rate r (concepts per unit time) with primes discovered at rate $q > 0$ (primes per unit time), the exploration ratio $\rho(t) \to 0$ as $t \to \infty$.*

Proof. At time t:

$$\rho(t) = \frac{|K(t)|}{|\mathcal{L}(t)|} \leq \frac{r \cdot t}{2^{k_0+qt}} \tag{6}$$

Since rt grows linearly and 2^{k_0+qt} grows exponentially, $\rho(t) \to 0$ by L'Hôpital's rule. □

4.4 Property 4: Non-prestatability

Theorem 11 (Future concepts are unknowable)**.** *Given known primes P and known concepts $K \subseteq \mathcal{L}(P)$, concepts involving an undiscovered prime p_{k+2} cannot be enumerated at the time p_{k+1} is discovered.*

Proof. The concept $p_{k+1} \cdot p_{k+2} \in \mathcal{L}(P \cup \{p_{k+1}, p_{k+2}\})$ but $p_{k+1} \cdot p_{k+2} \notin \mathcal{L}(P \cup \{p_{k+1}\})$ because $p_{k+2} \notin P \cup \{p_{k+1}\}$. The identity of p_{k+2} is unknown at p_{k+1}'s discovery. Therefore the full future space cannot be prestated from the current state. □

Remark 12 (Conservative non-prestatability)**.** *In Kauffman's chemical setting, non-prestatability is stronger: even the* types of reactions *possible with new molecules may be unknown. In the prime lattice, the operation is always multiplication—only the identity of future primes is unknown. This makes the prime version a conservative model: the weakest form that still satisfies the property. This is an advantage for formalization, as it isolates non-prestatability from operational novelty.*

Remark 13 (Empirical support)**.** *Random walks on the Boolean lattice of the first 20 primes (2^{20} nodes), using additive reinforcement with novelty parameter $\nu = 5.0$, produce a near-exact Zipf law: $\alpha_Z = 0.995$, $R^2 = 0.966$ (500 independent walks of 10,000 steps each; data from the empirical validation of the prime algebra [Ornelas Brand, 2026b]).*

5 Operations Before Collapse

In a pre-logical state, the standard connectives $(\wedge, \vee, \neg, \to)$ are undefined because they require truth values. But four operations remain well-defined:

Proposition 14 (Expansion)**.** *The operation* expand $: \mathcal{L}(P) \to \mathcal{L}(P \cup \{p_{k+1}\})$ *that adds a new prime to the repertoire is well-defined in a pre-logical state. It increases i (more possibilities open), does not require r, and doubles the cardinality of the possibility space (Theorem 7).*

Proof. The expansion operation acts on the lattice structure, not on truth values. It extends the prime set P to $P' = P \cup \{p_{k+1}\}$, creating $|\mathcal{L}(P)|$ new elements. No truth assignment is consulted. □

Proposition 15 (Partial evaluation)**.** *The operation $j_\epsilon : \hat{V} \to \hat{V}$ that increments the verification axis by $\epsilon > 0$ without resolving r is well-defined. It represents gathering evidence that narrows the possibility space without collapsing it:*

$$j_\epsilon(?, i, j, k) = (?,\ i - \delta,\ j + \epsilon,\ k) \tag{7}$$

where $\delta \geq 0$ is the reduction in potentiality induced by the evidence (some possibilities are eliminated).

Proof. The operation modifies only the i and j coordinates, which are defined in pre-logical states. The constraint $i > 0$ is maintained as long as $\delta < i$; the constraint $j < 1$ is maintained as long as $\epsilon < 1 - j$. Neither condition references r. □

Proposition 16 (Minimax preparation)**.** *Given candidate collapse outcomes $T = \{v_1, \ldots, v_m\} \subset V$, the **minimax preparation** is the point $v^* \in V$ that minimizes the maximum distance to all possible outcomes:*

$$v^* = \arg\min_{v \in V} \max_{v_\ell \in T} \|v - v_\ell\| \tag{8}$$

where $\|\cdot\|$ is the Euclidean norm on $V = [0,1] \times [-1,1]^3 \subset \mathbb{R}^4$. This operation is well-defined in a pre-logical state because it operates on the geometry of V, not on a specific truth value.

Proof. The Euclidean distance is continuous on V, and V is compact (closed and bounded in $\mathbb{R}^4$). The function $v \mapsto \max_\ell \|v - v_\ell\|$ is continuous as the maximum of finitely many continuous functions. Existence of v^* follows from the extreme value theorem. The computation requires only the candidates T (which the agent can enumerate from the possibility lattice), not the actual truth value. □

Proposition 17 (Selection)**.** *The operation $k_\delta : \hat{V} \to \hat{V}$ that commits to a direction reduces the effective possibility space by pruning branches of the lattice:*

$$k_\delta(?, i, j, k) = (?,\ i - \delta',\ j,\ k + \delta) \tag{9}$$

where $\delta' \geq 0$ is the reduction in potentiality caused by the commitment. Selection differs from partial evaluation: it reduces possibilities by choice *(closing doors), not by* evidence *(ruling out options).*

Proof. The quaternionic relation $ij = k$ ensures that selection is defined whenever both potentiality ($i > 0$) and some verification ($j > 0$ or $j = 0$ with prior evidence) are present. Axiom K3 of the quaternionic framework [Ornelas Brand, 2026c] states that $i > 0 \wedge j > 0$ enables but does not determine k: selection requires an agent's commitment beyond what the evidence provides. □

Remark 18 (Ordering of operations)**.** *The four operations have a natural partial order. Expansion (i increases) typically precedes partial evaluation (j increases, i decreases), which typically precedes selection (k increases, i decreases). But this ordering is not strict: an agent may select (k) before fully evaluating (j), accepting higher risk for faster action. The firefighter who brings all equipment is performing minimax; the one who sends a drone first is performing partial evaluation; the one who picks equipment based on experience is selecting.*

Remark 19 (Operations as structural types)**.** *The four operations specify what* type *of action is available in a pre-logical state: expanding possibilities, gathering evidence, preparing for all outcomes, or committing. The parameters δ (potentiality reduction from evidence) and δ' (potentiality reduction from commitment) are domain-specific: how much one drone observation narrows the firefighter's possibility space is an empirical question, not an algebraic one. What is universal is the structure: that these four operations are well-defined before truth exists, that they preserve the pre-logical constraints ($i > 0$, $j < 1$), and that they move the state toward collapse.*

6 Collapse as Algebraic Operation

Definition 20 (Collapse)**.** *The **collapse operator** $\mathbf{j} : \hat{V} \to V$ is the map that assigns a truth value to a pre-logical state:*

$$\mathbf{j}(?, i, j, k) = (r,\ i',\ 1,\ k') \tag{10}$$

where $r \in [0,1]$ is the crystallized truth, $i' \leq i$ (potentiality is reduced), $j = 1$ (verification is complete), and $k' \leq k$ (selection narrows to the chosen path). We write $\hat{V}_{\text{pre}}$ for the subset of $\hat{V}$ with $r = ?$.

The collapse is not instantaneous in general. We distinguish three stages:

Definition 21 (Stages of collapse)**.**

1. ***Pre-collapse**: $(?,\ i,\ j_0,\ k_0)$ with $j_0 \approx 0$. Possibilities are open, no evidence gathered.*
2. ***Partial collapse**: $(\tilde{r},\ i',\ j',\ k')$ with $0 < j' < 1$. Some evidence has been gathered; truth is partially crystallized ($\tilde{r}$ is a provisional estimate). The state has entered V but verification is incomplete.*
3. ***Total collapse**: $(r,\ 0,\ 1,\ 0)$. Truth is fully crystallized, no potentiality remains, verification is complete, selection is resolved. The state is fully logical.*

Theorem 22 (Properties of collapse)**.** *The collapse operator $\mathbf{j}$ satisfies:*

1. ***Irreversibility**: Once r is assigned, it cannot be retracted to $?$. The map $\mathbf{j}$ has no left inverse on its image.*
2. ***Monotonicity of verification**: Along any collapse trajectory, $j(t)$ is non-decreasing.*
3. ***Reduction of potentiality**: Along any collapse trajectory, $|i(t)|$ is non-increasing.*
4. ***Idempotence on logical states**: If $v \in V$ (already logical), then $\mathbf{j}(v) = v$. Collapse of a collapsed state is the identity.*

Proof. (1) By construction: $? \notin [0,1]$, so no element of V maps back to $\hat{V}_{\text{pre}}$. (2) Verification accumulates evidence; the constraint $j' \leq j''$ for $t' < t''$ follows from the fact that each partial evaluation adds $\epsilon > 0$ (Proposition 15). (3) Each partial evaluation reduces potentiality by $\delta \geq 0$, and each selection reduces it by $\delta' \geq 0$; the total is non-increasing. (4) If $r \in [0,1]$, the state is already in V, and the collapse operator acts as the identity on its codomain by definition. □

Remark 23 (Collapse is parametrised, not predicted)**.** *The collapse operator $\mathbf{j}$ does not predict* which *truth value r crystallises: the attractor μ_0 in Theorem 24 is a parameter determined by the domain—what the highway object is, what the judge rules, what the city builds. The framework characterises the universal* dynamics *of collapse (damped oscillation toward μ_0) and the universal* structure *of the pre-logical regime (Boolean lattice, four operations), while leaving the domain-specific* content *(which particular r emerges) to the domain. This separation is deliberate: the substrate independence of Section 10 depends on the process being invariant while the outcome varies.*

7 The Damped Wave as Free Energy Minimization

The transition from pre-logical to logical is not merely algebraic—it has dynamics. We prove that the collapse trajectory is governed by a damped harmonic oscillator that emerges exactly from the free energy principle.

Theorem 24 (DHO = FEP)**.** *Under the Laplace approximation with generalized coordinates $\tilde{\mu} = (\mu, \mu')$ and quadratic free energy*

$$F(\mu, \mu') = \tfrac{1}{2}\kappa(\mu - \mu_0)^2 + \tfrac{1}{2}\rho(\mu')^2 \tag{11}$$

the Friston dynamics (1) with dissipation $\Gamma = \operatorname{diag}(0, \gamma)$ and solenoidal coupling $Q = \left(\begin{smallmatrix} 0 & -\sigma \\ \sigma & 0 \end{smallmatrix}\right)$ yield:

$$\ddot{\mu} + \gamma\rho\,\dot{\mu} + \sigma\kappa(1 + \sigma\rho)\,(\mu - \mu_0) = 0 \tag{12}$$

whose underdamped solution ($\sigma\kappa(1 + \sigma\rho) > (\gamma\rho)^2/4$) is:

$$\mu(t) = Ae^{-dt}\sin(\omega_d\, t + \phi) + \mu_0 \tag{13}$$

with $d = \gamma\rho/2$ and $\omega_d = \sqrt{\sigma\kappa(1 + \sigma\rho) - (\gamma\rho)^2/4}$.

Proof. The free energy gradient is $\nabla_{\tilde{\mu}} F = \big(\kappa(\mu - \mu_0),\ \rho\mu'\big)^\top$. The shift operator and flow matrices are:

$$D = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \Gamma + Q = \begin{pmatrix} 0 & -\sigma \\ \sigma & \gamma \end{pmatrix}$$

Applying Eq. (1):

$$\dot{\mu} = \mu' - (-\sigma\rho\mu') = (1 + \sigma\rho)\,\mu' \tag{14}$$

$$\dot{\mu}' = -\sigma\kappa(\mu - \mu_0) - \gamma\rho\,\mu' \tag{15}$$

From (14): $\mu' = \dot{\mu}/(1 + \sigma\rho)$. Differentiating: $\dot{\mu}' = \ddot{\mu}/(1 + \sigma\rho)$. Substituting into (15) and multiplying by $(1 + \sigma\rho)$:

$$\ddot{\mu} + \gamma\rho\,\dot{\mu} + \sigma\kappa(1 + \sigma\rho)\,(\mu - \mu_0) = 0 \tag{16}$$

This is a damped harmonic oscillator with effective damping $b_{\text{eff}} = \gamma\rho$ and effective spring constant $k_{\text{eff}} = \sigma\kappa(1 + \sigma\rho)$. The characteristic equation $\lambda^2 + \gamma\rho\,\lambda + \sigma\kappa(1 + \sigma\rho) = 0$ has discriminant:

$$\Delta = (\gamma\rho)^2 - 4\sigma\kappa(1 + \sigma\rho) \tag{17}$$

When $\Delta < 0$, the roots are complex conjugates $\lambda = -d \pm i\omega_d$, yielding the underdamped solution (13). □

Remark 25 (Necessity of solenoidal coupling)**.** *With $Q = 0$ (purely dissipative dynamics) and $\Gamma = I$, the same quadratic F yields $\ddot{\mu} + (\kappa + \rho)\dot{\mu} + \rho\kappa(\mu - \mu_0) = 0$, whose discriminant $(\kappa - \rho)^2 \geq 0$ is always non-negative. This system is* always *overdamped—it can never oscillate. Solenoidal coupling ($\sigma > 0$) is therefore necessary for the underdamped regime within the FEP framework. Physically: the solenoidal component represents circulation in phase space—the system does not simply descend the free energy gradient, it* spirals *around the attractor while descending.*

Remark 26 (Interpretation for collapse dynamics)**.** *In the pre-logical context, the FEP parameters receive a natural interpretation:*

- *μ: the agent's current estimate (partial collapse state).*
- *μ_0: the attractor—the truth that will eventually crystallize.*
- *$A = D_{KL}(q_0 \| p)$: initial divergence between belief and reality.*
- *$d = \gamma\rho/2$: rate of surprise reduction.*
- *ω_d: frequency of oscillation between over- and under-estimates.*
- *σ: strength of the solenoidal coupling that enables oscillation.*

Collapse is not a sudden jump but a damped oscillation*: the agent overshoots, corrects, overshoots less, corrects again, spiraling toward the truth.*

7.1 Parameter mapping

Table 1: Parameter mapping between the wave model and the FEP. Each row is a definitional identification under the derivation, not a loose analogy.

Wave	**FEP equivalent**	**Interpretation**
A (amplitude)	$D_{KL}(q_0 \| p)$	Initial divergence between belief and reality
d (damping)	$\gamma\rho/2$	Dissipative precision on velocity; rate of surprise reduction
f (frequency)	$\frac{1}{2\pi}\sqrt{\sigma\kappa(1+\sigma\rho) - (\gamma\rho)^2/4}$	Solenoidal–curvature interaction
ϕ (phase)	Prior mean offset	Initial belief about the state
μ_0 (zero line)	Attractor	The state the system expects to occupy
Ae^{-dt} (envelope)	$F(t) \to F_{\min}$	Free energy approaching its minimum
σ (new)	Solenoidal coupling	Circulation in phase space; enables oscillation

7.2 Conditions for oscillation

The system oscillates when:

$$\sigma\kappa(1+\sigma\rho) > \frac{(\gamma\rho)^2}{4} \tag{18}$$

Corollary 27 (Three dynamical regimes)**.**

1. ***Underdamped*** *($\sigma\kappa(1+\sigma\rho) > (\gamma\rho)^2/4$): Oscillation. The system spirals around the attractor with decaying amplitude.*
2. ***Critically damped*** *($\sigma\kappa(1+\sigma\rho) = (\gamma\rho)^2/4$): Fastest convergence without oscillation.*
3. ***Overdamped*** *($\sigma\kappa(1+\sigma\rho) < (\gamma\rho)^2/4$): Monotone exponential decay. No oscillation.*

8 Two Routes to Oscillation

Proposition 28 (Geometric route to oscillation)**.** *Under the standard Friston dynamics ($\Gamma = I$, $Q = 0$) with a* non-diagonal *free energy that includes a cross-term η:*

$$F_\eta(\mu, \mu') = \tfrac{1}{2}\kappa(\mu - \mu_0)^2 + \tfrac{1}{2}\rho(\mu')^2 + \eta(\mu - \mu_0)\mu' \tag{19}$$

the dynamics yield:

$$\ddot{\mu} + (\kappa + \rho)\,\dot{\mu} + [\rho\kappa + \eta(1-\eta)]\,(\mu - \mu_0) = 0 \tag{20}$$

with discriminant $\Delta = (\kappa - \rho)^2 - 4\eta(1-\eta)$. For $0 < \eta < 1$ and $(\kappa - \rho)^2 < 4\eta(1-\eta)$, the system is underdamped—oscillation arises without *solenoidal coupling.*

Proof. The gradient is $\nabla F_\eta = \big(\kappa(\mu - \mu_0) + \eta\mu',\ \rho\mu' + \eta(\mu - \mu_0)\big)^\top$. Standard Friston dynamics ($\dot{\tilde{\mu}} = D\tilde{\mu} - \nabla F_\eta$):

$$\dot{\mu} = (1-\eta)\,\mu' - \kappa(\mu - \mu_0) \tag{21}$$

$$\dot{\mu}' = -\rho\,\mu' - \eta(\mu - \mu_0) \tag{22}$$

From the first equation: $\mu' = (\dot{\mu} + \kappa(\mu - \mu_0))/(1-\eta)$. Differentiating and substituting:

$$\ddot{\mu} + (\kappa + \rho)\,\dot{\mu} + [\rho\kappa + \eta(1-\eta)]\,(\mu - \mu_0) = 0$$

The discriminant is $(\kappa+\rho)^2 - 4[\rho\kappa + \eta(1-\eta)] = (\kappa-\rho)^2 - 4\eta(1-\eta)$, which is negative when $(\kappa-\rho)^2 < 4\eta(1-\eta)$. □

Remark 29 (Two independent routes)**.** *Theorem 24 and Proposition 28 provide two independent routes to oscillation within the FEP:*

1. ***Dynamical route**: diagonal F with solenoidal coupling ($\sigma > 0$). The system spirals due to non-dissipative flow.*
2. ***Geometric route**: non-diagonal F with standard dynamics ($Q = 0$). The free energy landscape itself drives oscillation.*

The geometric route is physically suggestive: the cross-term η represents coupling between a system's state and its rate of change within the free energy functional. In the pre-logical context, this means that what the agent currently believes *and* how fast that belief is changing *are coupled within the free energy landscape —the shape of uncertainty itself can drive oscillatory collapse.*

9 Information as Compound Concept

The duality synthesis framework [Ornelas Brand, 2026d] shows that opposing poles generate a compound concept through $a \oplus \bar{a} = \mathrm{lcm}(\Phi(a), \Phi(\bar{a}))$. We apply this to the fundamental duality of this paper.

Proposition 30 (Information as compound concept)**.** *The duality (pre-logical, logical) generates **information** as its compound concept. Specifically:*

1. *The two poles are genuinely opposing: the pre-logical state $(?, i, j_0, k_0)$ has maximal potentiality and no truth; the logical state $(r, 0, 1, 0)$ has crystallised truth and no potentiality. Neither is reducible to the other.*
2. *Information requires* both *poles: it is neither pure possibility (that is ignorance) nor pure truth (that is fact), but the* reduction *of possibility* toward *truth.*
3. *The verification axis j measures the degree of collapse between the poles. The selection axis k encodes the agent's commitment.*

Proof. This is a structural argument, not a computation in the prime algebra: the pre-logical/logical duality is a *meta*-duality (a duality of the framework itself, not of concepts within the 72-primitive ontology), so the prime signatures Φ(pre-logical) and Φ(logical) are not defined. What we verify is the *pattern* of synthesis from Ornelas Brand [2026d].

Opposition. The pre-logical state has $r = ?$ (undefined), $i > 0$ (open possibilities), $j < 1$ (incomplete verification). The logical state has $r \in [0, 1]$ (crystallised), $i = 0$ (no remaining possibilities), $j = 1$ (complete verification). On every axis where one pole is maximal, the other is minimal. This is the strongest form of opposition in the quaternionic framework.

Irreducibility. Pure possibility (r absent) cannot derive truth; pure truth ($i = 0$) cannot recover the space of alternatives that was collapsed. The concept "information" requires reference to both: information is the measure of how much possibility was eliminated to reach truth.

Compatibility with Shannon. Shannon entropy $H = -\sum p_i \log p_i$ [Shannon, 1948] measures information within the r axis: probabilities are truth values, and $I = H_{\text{before}} - H_{\text{after}}$ is the reduction from uncertainty to certainty. This is the restriction of the pre-logical/logical transition to the regime where r is defined (indeterminacy, not indefiniteness). The quaternionic framework does not derive Shannon entropy; it extends the concept to the regime where r is not yet defined, with geodesic distance on S^3 replacing entropy as the measure of separation between current state and attractor. □

Remark 31 (The bit as elementary collapse)**.** *A single bit of information corresponds to one step of collapse: the resolution of one binary question ("is the object a tree? yes/no"). In the prime lattice, this is a Hamming-distance-1 move. In the FEP dynamics, it is one half-period of the damped oscillation (overshoot, then correction). The bit is the quantum of the pre-logical-to-logical transition.*

10 Substrate Independence

The structure (pre-logical state → collapse → logical state) is substrate-independent: the same algebraic framework applies across domains that differ in their physical instantiation. We demonstrate this with four examples.

10.1 Quantum measurement

Domain: Quantum physics — Schrödinger's cat.

Component	Instantiation
Pre-logical state	Closed box; Re undefined
Potentiality (i)	Alive and dead as superposed possibilities
Verification (j)	Opening the box / measuring
Selection (k)	Design of the experiment
Collapse	Measurement → eigenvalue
Logical state	Alive or dead—crystallized by observation

The quantum case is often presented as exotic. It is not. The firefighter facing an unidentified highway obstruction is in the same algebraic situation: a superposition of possibilities that collapses upon observation. The difference is substrate (quantum amplitudes vs. macroscopic objects), not structure.

10.2 Legal reasoning

Domain: Law — Case without precedent.

Component	Instantiation
Pre-logical state	Novel case; no applicable jurisprudence
Potentiality (i)	Possible legal arguments
Verification (j)	Judicial ruling
Selection (k)	Lawyer's litigation strategy
Collapse	Ruling → new precedent
Logical state	Jurisprudence created—crystallized by the court

The legal method IRAC (Issue, Rule, Application, Conclusion) maps directly onto the pre-logical framework: *Issue* = identify which Re does not yet exist (the pre-logical state); *Rule* = existing Re (crystallized law); *Application* = $j + k$ (verify applicability, select strategy); *Conclusion* = collapse → new Re.

Each law can be viewed as a quaternion: r is the literal text, i the possible interpretations, j the case law that verifies it, k the selection of which precedent applies. Changing jurisdiction (e.g., from one state to another) is a rotation q on S^3: the commutation theorem predicts that laws depending only on Re (constitutional provisions) are invariant under rotation, while laws depending on Im (local interpretation) vary.

10.3 Emergency response

Domain: Emergencies — Unidentified highway obstruction.

Component	Instantiation
Pre-logical state	Vague report: "large object on Highway 45"
Potentiality (i)	Tree, vehicle, animal, boulder
Verification (j)	Arriving at the scene
Selection (k)	Equipment choice
Collapse	Visual identification → confirmed object
Logical state	"It is a fallen tree"—crystallized by observation

The optimal decision under a pre-logical state is the minimax preparation (Proposition 16): carry equipment that minimizes the maximum distance to all possible outcomes. The firefighter brings crane, hammers, *and* winch because, regardless of what the object is, the preparation should be close to the solution.

Partial evaluation is available: send a drone first (j partial, low cost) before committing the full crew (j total, high cost). The sequence of partial collapses is an optimization over the lattice of possibilities.

10.4 Urban planning

Domain: Urbanism — City 2040.

Component	Instantiation
Pre-logical state	Uncertain future: water, transit, growth
Potentiality (i)	Scenarios for population, climate, economy
Verification (j)	Construction (irreversible)
Selection (k)	Budget allocation
Collapse	Building → physical infrastructure
Logical state	City built—crystallized by construction

Each urban decision collapses possibilities: building a highway *here* eliminates a park *there*. Each j that is executed closes branches of the adjacent possible. The lattice formalism (Section 4) makes this precise: the known set K grows, but the adjacent possible $AP(K)$ changes structure with each commitment.

The irreversibility of construction makes urban planning a particularly pure example of collapse: unlike quantum measurement (which can be repeated) or legal rulings (which can be overturned on appeal), a building *cannot be unbuilt* at reasonable cost.

10.5 Structural invariance

Table 2: Substrate independence: the same structure across four domains.

	Quantum	**Legal**	**Emergency**	**Urban**
r undefined	Superposition	No precedent	Unknown object	Unknown future
$i > 0$	Amplitudes	Arguments	Possibilities	Scenarios
j (collapse)	Measurement	Ruling	Observation	Construction
k (selection)	Experiment	Strategy	Equipment	Budget
Lattice	Hilbert space	Case space	Object space	Scenario space
Irreversibility	Wavefunction	Partial	Low	Total

The structural invariance across these four domains—differing in substrate, time scale, reversibility, and social context—supports the claim that pre-logical states are not a special case but the *generic* situation for reasoning agents. Logic (the assignment and manipulation of truth values) is the endpoint of a process that begins in the pre-logical regime.

11 Discussion

11.1 Step zero

Classical epistemology describes a chain: (1) opposites exist (hot/cold), (2) a scale connects them, (3) the zero point is arbitrary (Kelvin, Fahrenheit, Celsius are rotations on S^3), (4) a compound concept emerges (temperature), (5) perspective and context rotate the reading.

The pre-logical state is **step zero**: before the agent knows *which* opposites apply. The firefighter does not know whether the problem is heavy/light, organic/inorganic, mobile/fixed. The lawyer does not know whether the case is civil/criminal, state/federal. The urban planner does not know whether the problem is water, drainage, or transit.

Verification (j) moves the agent from step zero to step one: collapsing not only truth but *which framework of opposites applies.* The pre-logical state is the pre-structure of structure—navigating the adjacent possible of dualities, not of concepts.

11.2 Relation to probability theory

Pre-logical states are not probabilistic states. Probability theory assigns a distribution $P(r)$ over truth values; it handles *indeterminacy* (the truth exists but is unknown). The pre-logical state handles *indefiniteness* (no truth has crystallized). The distinction matters:

- Under indeterminacy, Bayesian updating is optimal.
- Under indefiniteness, the optimal strategy is minimax (Proposition 16)—minimize worst-case distance, not expected distance.

The two regimes connect through the FEP: Bayesian updating corresponds to the underdamped and overdamped regimes (convergence to an existing attractor), while transitions between attractors correspond to shifts in the pre-logical landscape—new primes entering the lattice.

11.3 Limitations

1. **Structural, not quantitative.** The substrate independence demonstrations establish that the same algebraic framework applies across four domains, but they do not derive numerical predictions for any specific domain. The value is in the structural correspondence: identifying which operations are available and which properties hold, not in predicting particular outcomes.
2. **Collapse endpoint is domain-specific.** The FEP dynamics describe *how* collapse occurs (damped oscillation toward the attractor μ_0) but the attractor itself is a parameter determined by the domain—what the object on the highway actually is, what the judge actually rules. This is by design: the framework characterises the universal *process* of collapse while leaving the domain-specific *content* to the domain.
3. **Four domains is few.** The substrate independence claim would be strengthened by additional domains (medicine, AI/ML, financial markets).
4. **Conservative lattice.** The prime product lattice allows all combinations; real domains have constraints. The lattice is the "ideal gas" of the adjacent possible—exact and tractable, but missing thermodynamic constraints.

12 Conclusion

Logic is not the beginning of reasoning—it is the end.

Before truth exists, there is navigation: possibilities that expand (Theorem 7), paths that depend on history (Theorem 9), exploration that falls exponentially behind the growth of the possible (Theorem 10), and futures that cannot be prestated (Theorem 11). The agent gathers evidence (partial evaluation), prepares for all outcomes (minimax preparation), and commits (selection)—all without a truth value to operate on.

The transition from this pre-logical regime to the logical one—collapse—is a damped oscillation governed by the free energy principle (Theorem 24), with solenoidal coupling necessary for the system to spiral rather than slide to its attractor (Remark 25). Two independent routes to oscillation exist: dynamical (solenoidal flow) and geometric (cross-coupled free energy) (Proposition 28).

The duality between pre-logical and logical generates information as its compound concept (Proposition 30): not possibility alone (that is ignorance) and not truth alone (that is fact), but the *reduction* of the one toward the other. Shannon entropy operates within the truth axis (where r is defined); the quaternionic framework extends the concept to the regime where r is not yet defined, with geodesic distance replacing entropy as the measure of separation.

The structure is substrate-independent: quantum measurement, legal reasoning, emergency response, and urban planning all instantiate the same algebraic framework with different physical substrates. This suggests that the pre-logical state is not an edge case but the generic condition of reasoning agents—and that classical logic describes the world after the measurement, not before it.

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